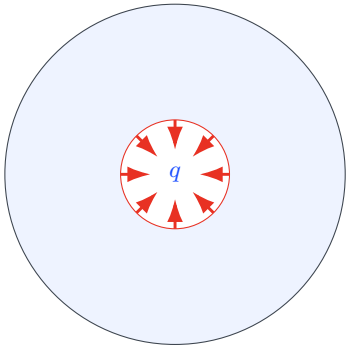
点电荷电势：原点外拉普拉斯为零，小球通量固定集中源强度



穿孔区域的内边界法向指向原点

$$r > 0 : \quad \nabla^2(1/r) = 0$$

$$\oint_{r=\varepsilon} \nabla\varphi \cdot \hat{\mathsf{n}}_{V_\varepsilon}\, dS = \frac{q}{\varepsilon_0}$$

$$\nabla^2\varphi = -\frac{q}{\varepsilon_0}\delta^{(3)}(\mathsf{r})$$

负号来自 $\mathsf{E} = -\nabla\varphi$。